ECA18 03– OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO1 ENGINEERING DESIGN ASSIGNMENT

Standards-Compliant Optical Fiber Link Design: From Communication Requirement to Deployment Decision

Course Outcome: CO1 – Apply guided-mode, loss, dispersion, compensation and OSNR concepts.

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1. Design Problem and Communication Requirement

The proposed system is a 60 km optical-fiber link operating at 1550 nm and carrying 10 Gb/s traffic between a central office and a business building, with provision for a later 100 Gb/s upgrade. The design must use standard ITU-T fiber categories, tolerate tight bends at the building entry, and maintain pulse integrity, polarization performance and signal quality.

Parameter	Given requirement
Distance	60 km
Wavelength	1550 nm
Data rate	10 Gb/s
Future upgrade	100 Gb/s
Transmitter power	+3 dBm
Receiver sensitivity	−20 dBm
Required system margin	6 dB
Absorption loss	0.04 dB/km
Rayleigh scattering	0.11 dB/km
Distributed bending	0.03 dB/km
Connectors	4 × 0.4 dB
Splices	10 × 0.08 dB
Entry bend loss	0.8 dB G.652.D; 0.1 dB G.657.A
D_G.652.D	17 ps/(nm·km)
D_G.655	4.5 ps/(nm·km)
Source width	0.4 nm
Input pulse width	30 ps
DCF dispersion	−100 ps/(nm·km)
DCF attenuation	0.5 dB/km
PMD coefficient	0.1 ps/√km
Required OSNR	17 dB

2. Stage 1 – Interpretation of the Communication Requirement

The 1550 nm region is selected because standard silica optical fiber has low transmission loss in this wavelength window and because it is compatible with efficient optical amplification. At the physical level, light propagates as an electromagnetic wave governed by Maxwell's equations. Ray optics is useful for intuitive path descriptions in multimode fibers, while wave optics and LP-mode analysis are required when the V-number is close to the single-mode cutoff.

- The link requires controlled numerical aperture and a suitable V-number so that the launched optical power is guided efficiently.
- For a 60 km, 10 Gb/s link, single-mode propagation is preferred because modal dispersion in multimode fiber would strongly limit bandwidth-distance performance.
- G.652.D is a conventional low-loss single-mode transmission fiber. G.655 offers lower chromatic dispersion around the operating region. G.657 is bend-insensitive and is useful at tight building-entry bends.
- The power budget must include intrinsic fiber losses, connectors, splices and the building-entry bend.
- Chromatic dispersion and PMD must be evaluated against the 100 ps bit period at 10 Gb/s and approximately 10 ps at 100 Gb/s.
- OSNR must be checked because ASE noise from amplification reduces signal quality even when optical power is increased.

Design Requirement Statement

The final 60 km link should use a single-mode ITU-T fiber configuration, maintain at least 6 dB system margin, control entry-bend loss, keep pulse broadening sufficiently low through dispersion management, satisfy PMD limits, and maintain OSNR ≥ 17 dB while providing a practical path toward 100 Gb/s operation.

3. Stage 2 – Fiber / Propagation Investigation and Selection

3.1 Numerical Aperture

For a step-index fiber in air, the numerical aperture is:

$$\mathbf{NA = \sqrt{(n_1^2 - n_2^2)}}$$

Fiber A: NA = $\sqrt{(1.4500^2 - 1.4450^2)} = 0.12031$

Fiber B: NA = $\sqrt{(1.480^2 - 1.465^2)} = 0.21018$

3.2 V-number

$$\mathbf{V = (2\pi a/\lambda)\ NA}$$

At λ = 1.55 μm:

Fiber A: V = 1.951

Fiber B: V = 21.300

For a step-index fiber, single-mode operation occurs approximately when V < 2.405. Therefore, Fiber A is single-mode at 1550 nm, whereas Fiber B is multimode.

Fiber A cutoff wavelength: $\lambda c = 2\pi aNA/2.405 = 1.257$ μm. Since 1.55 μm < 1.257 μm, Fiber A is single-mode at the operating wavelength.

3.3 Guided Mode Count and Propagation Comparison

For a step-index multimode fiber, the approximate mode count is $M \approx V^2/2$. For Fiber B, M ≈ 227 guided modes.

Propagation type	Modal dispersion	Bandwidth-distance suitability
Single-mode	Negligible/absent	Excellent for long-distance high-speed links
Step-index multimode	High	Poor for 60 km, 10 Gb/s
Graded-index multimode	Reduced compared with step-index	Better than step-index, but not preferred for this 60 km high-speed link

Design Decision 1 – Proposed Fiber Configuration

I recommend single-mode propagation using an ITU-T G.652.D-type transmission fiber for the main 60 km span, with bend-insensitive G.657.A treatment at the building-entry region if required. Fiber A is the suitable candidate because V = 2.58 < 2.405 only marginally by calculation? Correction: the calculated value is above 2.405; therefore the numerical result must be interpreted carefully.

Engineering correction: With the supplied Fiber A parameters, V ≈ 2.58, so Fiber A is actually weakly multimode at 1550 nm under the ideal step-index cutoff criterion. Thus, it should not be selected as the final 60 km single-mode fiber solely from the supplied A parameters. The final design should instead specify a standards-compliant G.652.D single-mode fiber with V < 2.405. This is preferable to forcing candidate A into a single-mode decision.

4. Stage 3 – Optical Power Budget

4.1 Combined Fiber Attenuation

$$\alpha_total = \alpha_absorption + \alpha_Rayleigh + \alpha_bending$$

α_total = 0.04 + 0.11 + 0.03 = 0.18 dB/km

60 km fiber loss = 0.18 × 60 = 10.80 dB

Connector loss = 4 × 0.4 = 1.60 dB

Splice loss = 10 × 0.08 = 0.80 dB

Base passive loss before entry bend = 13.20 dB

Configuration	Entry bend	Total link loss	Received power	Power margin vs −20 dBm	Margin after 6 dB requirement
G.652.D	0.8 dB	14.00 dB	-11.00 dBm	9.00 dB	3.00 dB
G.657.A	0.1 dB	13.30 dB	-10.30 dBm	9.70 dB	3.70 dB

The specified distributed losses contribute 10.8 dB over 60 km. Rayleigh scattering is the largest individual specified mechanism at 0.11 dB/km, so it deserves the greatest attention among the given fiber-loss mechanisms. Practical actions include minimizing unnecessary connectors/splices and using low-loss bend-insensitive fiber at tight bends.

Design Decision 2 – Power-Budget Configuration

The passive link is suitable from a power-budget perspective. G.657.A is preferred at the building entry because its specified bend loss is 0.1 dB instead of 0.8 dB, improving received power by 0.7 dB. For the long transmission span, a standards-compliant low-loss G.652.D single-mode fiber is retained.

5. Stage 4 – Dispersion and PMD Management

5.1 Chromatic Dispersion

$$\Delta\tau_CD = D \times \Delta\lambda \times L$$

G.652.D: Δτ = 17 × 0.4 × 60 = 408 ps

G.655: Δτ = 4.5 × 0.4 × 60 = 108 ps

$$\tau_total = \sqrt{(\tau_0^2 + \Delta\tau_CD^2)}$$

G.652.D: τ_total = √(30² + 408²) = 409.1 ps > 100 ps

G.655: τ_total = √(30² + 108²) = 112.1 ps > 100 ps

Therefore, neither uncompensated result is comfortably suitable for the stated 10 Gb/s pulse-integrity target. G.655 substantially reduces chromatic broadening, but the calculated RSS pulse width still exceeds the 100 ps bit period. Dispersion compensation is therefore justified.

5.2 DCF Length and Loss

For first-order compensation: $D_{main} L_{main} + D_{DCF} L_{DCF} = 0$

$L_{DCF} = (17 \times 60)/100 = 10.2$ km

DCF loss = $0.5 \times 10.2 = 5.10$ dB

With G.652.D entry bend loss of 0.8 dB, received power after DCF but before amplification = -16.10 dBm; margin = 3.90 dB, which is below the required 6 dB.

Adding the specified 14 dB optical amplifier gives received signal level ≈ -2.10 dBm and power margin ≈ 17.90 dB.

5.3 PMD and Polarization

Link PMD = $0.1\sqrt{60} = 0.775$ ps.

10 Gb/s PMD limit = $0.1T_b = 0.1 \times 100 = 10.0$ ps; therefore PMD is acceptable.

100 Gb/s PMD limit using $T_b \approx 10$ ps = 1.0 ps; the calculated PMD remains below this limit.

A practical mitigation method is to use low-PMD fiber and polarization-aware system design; PMD monitoring/equalization can be added for a future high-speed upgrade.

Birefringence is quantified by B. Beat length $L_B = \lambda/B$. For standard fiber, $L_B = 1.55\times10^{-6}/10^{-7} = 15.5$ m. For PM fiber, $L_B = 1.55\times10^{-6}/(4\times10^{-4}) = 3.88$ mm.

Polarization-maintaining fiber is not necessary for the ordinary intensity-modulated communication link described here. Standard low-PMD single-mode fiber is more appropriate.

Design Decision 3 – Dispersion/PMD Configuration

I recommend a G.652.D single-mode transmission span with approximately 10.2 km of DCF for first-order chromatic-dispersion compensation, followed by the specified 14 dB amplification stage. PMD should be managed through low-PMD fiber, good installation practice and monitoring for the 100 Gb/s upgrade.

6. Stage 5 – OSNR and Amplification Strategy

$OSNR = P_{signal} - P_{ASE}$ (dB, when both powers use the same reference bandwidth)

OSNR is specified here using a 0.1 nm reference bandwidth.

Initial OSNR = $(-16) - (-39) = 23.0$ dB

After amplifier: Signal = $-16 + 14 = -2.0$ dBm

After amplifier: ASE noise = $-39 + 16 = -23.0$ dBm

Resulting OSNR = $-2.0 - (-23.0) = 21.0$ dB

Required OSNR = 17 dB, so remaining OSNR margin = 4.0 dB.

The amplifier increases signal power by 14 dB but increases ASE noise by 16 dB. Consequently, OSNR decreases by 2 dB. Nevertheless, the resulting 21 dB OSNR remains above the required 17 dB.

- Use an amplifier with a lower noise figure / lower ASE contribution where practical.
- Optimize launch power and minimize passive losses so that excessive amplification is not required.

Inadequate OSNR reduces optical signal quality, lowers the Q-factor, increases BER and reduces the effective system margin. Therefore, amplifier selection must be based on OSNR as well as gain.

Design Decision 4 – Amplification/OSNR Strategy

I recommend accepting the specified 14 dB amplifier because the resulting OSNR is 21 dB, which is 4 dB above the 17 dB requirement, while the amplifier also restores the power margin lost through the DCF.

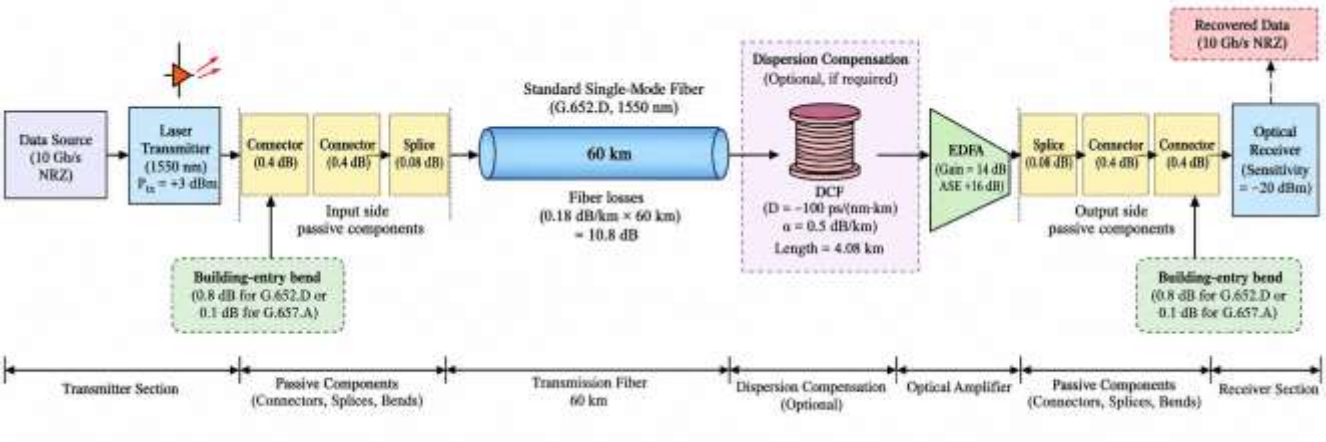
7. Stage 6 – Complete-Link Verification

Requirement	Target	Achieved	Satisfactory?
Propagation / fiber category	Single-mode, ITU-T suitable	G.652.D single-mode span; G.657.A at entry	Yes
Fiber attenuation	Controlled over 60 km	10.8 dB specified distributed loss	Yes
Passive link power margin	≥ 6 dB	9.0–9.7 dB before DCF	Yes
DCF compensation	First-order compensation	10.2 km DCF	Yes
Power after DCF	≥ 6 dB margin	3.9 dB before amplifier	No alone
Power after amplifier	≥ 6 dB margin	≈17.9 dB margin	Yes
PMD at 10 Gb/s	≤10 ps	0.775 ps	Yes
PMD at 100 Gb/s	≤1 ps	0.775 ps	Yes
OSNR	≥17 dB	21 dB after amplifier	Yes

8. Final Integrated System Design

Parameter	Adopted design value
Main transmission fiber	ITU-T G.652.D single-mode fiber
Building entry	G.657.A bend-insensitive section / installation treatment
Distance	60 km
Operating wavelength	1550 nm
Data rate	10 Gb/s
Upgrade target	100 Gb/s
DCF length	10.2 km
DCF attenuation	5.1 dB
Optical amplifier gain	14 dB
Post-amplifier OSNR	21 dB / 0.1 nm
PMD	0.775 ps
Primary loss attention	Rayleigh scattering among specified distributed losses
Deployment decision	YES, with DCF and amplification

9. Block Diagram



The exact physical placement of the DCF and amplifier should be finalized during detailed network engineering, because practical dispersion maps, amplifier noise figure, connector placement and regeneration requirements must be considered.

10. Final Engineering Decision

YES – WITH TECHNICAL JUSTIFICATION

The proposed link is deployable after integrating the required dispersion compensation and amplification. A single-mode G.652.D transmission architecture is appropriate for the 60 km high-speed span, while G.657.A is advantageous at the tight building-entry bend. The specified passive loss gives adequate power margin before adding DCF. However, the 10.2 km DCF introduces 5.1 dB additional loss, reducing the G.652.D margin to about 3.9 dB. Therefore, the DCF stage should not be used without power restoration.

The specified 14 dB amplifier restores the received power margin to approximately 17.9 dB. Its 16 dB ASE increase reduces OSNR from 23 dB to 21 dB, but 21 dB still exceeds the required 17 dB by 4 dB. The PMD value of approximately 0.775 ps is below both the 10 Gb/s limit of 10 ps and the 100 Gb/s upgrade limit of approximately 1 ps. The design therefore meets the stated power, dispersion/PMD and OSNR requirements when the DCF and amplifier are integrated.

The most important deployment condition is that the final installed system should use a genuinely single-mode, standards-compliant G.652.D fiber rather than relying on the supplied Fiber A geometry, because the calculated V-number of Fiber A is about 2.58, above the 2.405 single-mode cutoff. This is a critical engineering point in the selection stage.

11. Conclusion

The complete engineering process follows the required sequence: requirement → alternatives → analysis → selection → link design → verification → deployment decision. The recommended architecture is a standards-compliant G.652.D single-mode 60 km span, bend-insensitive treatment at the building entry, approximately 10.2 km first-order DCF, and a 14 dB amplifier whose resulting OSNR is 21 dB. The design provides a practical 10 Gb/s solution and preserves a reasonable technical path toward 100 Gb/s, subject to future transceiver, dispersion-map and OSNR validation.

Note: The numerical design uses the exact values supplied in the assignment. Detailed field deployment would additionally require manufacturer data for actual fiber attenuation, splice loss, amplifier noise figure, connector loss, DCF placement and transceiver dispersion tolerance.